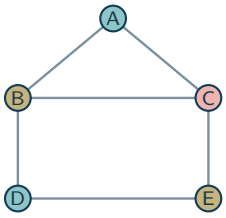
GRAPH THEORY LADDER 7 OF 9

Colouring and bipartite graphs

Neighbours get different colours, and two colours suffice exactly when there is no odd cycle.

1 / THE RULE AND THE PICTURE

The chromatic number is the fewest colours that keep every edge two-coloured



house: 3 colours



$K_{3,3}$: 2 colours

A proper colouring gives adjacent vertices different colours, and the smallest number needed is the chromatic number χ. A graph can be coloured with two colours exactly when it is bipartite:

$$\chi \leq 2 \iff \text{no odd cycle}$$

The house needs 3 because of its triangle, and $K_{3,3}$ needs only 2.

2 / KNOWN VALUES

The chromatic number of familiar graphs

graph	χ	reason
path P_n, $n \geq 2$	2	alternate the colours
even cycle C_{2m}	2	alternate round the ring
odd cycle C_{2m+1}	3	two colours fail at the closing edge
complete graph K_n	n	every pair adjacent
complete bipartite $K_{m,n}$	2	one colour for each side
house graph	3	the triangle needs three
Petersen graph	3	two colours fail, three succeed
any planar graph	at most 4	the four colour theorem

3 / COUNTING COLOURINGS

The chromatic polynomial counts them for every number of colours

Colouring the triangle ABC first, then D and E, shows that the house has $k(k-1)(k-2)\big[(k-1)+(k-2)^2\big]$ proper colourings with k colours, which is 0 for $k=2$, 18 for $k=3$ and 168 for $k=4$, and an exhaustive count confirms both numbers.
The 5-cycle has $(k-1)^5-(k-1)$, giving 0 for $k=2$ and 30 for $k=3$, in agreement with the rule that an odd cycle needs three colours.

4 / A SIMPLE METHOD AND ITS LIMIT

Greedy colouring uses at most $\Delta+1$ colours

Colour the vertices in any order, giving each the first colour not already used by a neighbour, and a vertex with Δ neighbours can never be forced beyond $\Delta+1$ colours, so the house with $\Delta=3$ needs at most 4.
The bound is not always tight, since $\chi=3$ here, and Brooks' theorem sharpens it to Δ for every connected graph except complete graphs and odd cycles.
Deciding whether a graph can be coloured with three colours is NP-complete, so no fast method is known.

5 / MAPS AND TIMETABLES

Colouring is scheduling in disguise

To colour a map, make each country a vertex and join countries that share a border, which gives a planar graph, and the four colour theorem of Appel and Haken (1976) says four colours always suffice.
For a timetable, make each exam a vertex, join two exams that share a student, and read the colours as time slots: exams with the house-graph pattern of clashes need three slots.

6 / TRY THESE

Decide the chromatic number

(a) What is $\chi(C_7)$? It is 3, since 7 is odd. **(b)** Can the cube graph be 2-coloured? Yes, by the parity of the number of 1s in the code, so it is bipartite. **(c)** Why does K_4 need four colours? Every vertex is adjacent to every other.

ASIDE / BIPARTITE MEANS TWO SIDES

Matching problems, such as pairing people with jobs, live on bipartite graphs, where Hall's marriage theorem tells when a perfect matching exists.

LINKS TO OTHER SHEETS

Planar graphs, where four colours suffice, are on the Euler formula sheet; the next sheet counts spanning trees with a matrix.

Convention. χ chromatic number; Δ the largest degree; a proper colouring gives adjacent vertices different colours; the house colouring in the picture uses 3 colours.